

Warm-up

Analyze the convergence of the following series.

$$(1) \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2}, \quad (2) \sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n}}, \quad (3) \sum_{n=2}^{\infty} (-1)^n n^2.$$

Medium

Analyze the convergence of the following series:

$$(1) \sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2 + 1}, \quad (2) \sum_{n=1}^{\infty} \frac{n \sin(n)}{n^3 + 2}, \quad (3) \sum_{n=1}^{\infty} \frac{\sin(\pi n) \sqrt{n}}{2n + 3}.$$

(4) Starting with a_1 , how many terms of the sequence $a_n = (-1)^n/n^2$ do you need to add before the alternating series error bound guarantees the partial sum is at most .01 away from the value of the infinite sum $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$?

Stretch

(1) Consider the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \dots$. This converges by the alternating series test, and we will show in a future video that it converges to $\ln(2)$. Shuffle the terms of the series into groups of three of the form

$$\left(\frac{1}{2k-1} - \frac{1}{4k-2} \right) - \frac{1}{4k}$$

so that the series now looks like

$$\left(1 - \frac{1}{2} \right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6} \right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10} \right) - \frac{1}{12} + \dots$$

First check that this hasn't changed which terms appear with which signs (do we still have all the odd denominators? Even denominators?) and then determine what this rearranged series converges to.

(2) Using the idea of (1), come up with a strategy to rearrange the terms to converge to whatever number you want. It may be helpful to draw some pictures.